

Problem 1. Let X and Y be two topological spaces and $f : X \rightarrow Y$.

- (i) Prove that f is continuous if and only if $f^{-1}(C)$ is closed for all closed sets C in Y .

Proof. ($\implies$) Suppose f is continuous. Let C be a closed set in Y . Our goal is to show that $f^{-1}(C)$ is closed in X . Since C is closed in Y , it follows that its complement C^c is open in Y . Since f is continuous, its preimage $f^{-1}(U)$ where $U = C^c$ is open in X . This tells us that $[f^{-1}(U)]^c$ is closed in X . Since $[f^{-1}(U)]^c = f^{-1}(U^c) = f^{-1}(C)$, it follows that $f^{-1}(C)$ is closed in X .

($\impliedby$) Suppose that for all closed sets C in Y , $f^{-1}(C)$ is closed in X . Our goal is to show that f is continuous; that is, we want to show that for all open sets U in Y , $f^{-1}(U)$ is open in X . To this end, let U be an open set in Y . This tells us that $U^c = V$ is a closed set in Y . Then $f^{-1}(V)$ is closed in X by assumption. That is, $[f^{-1}(V)]^c$ is open in X . Since

$$[f^{-1}(V)]^c = f^{-1}(V^c) = f^{-1}(U),$$

it follows that $f^{-1}(U)$ is open in X which is our desired result. ■

- (ii) Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for all $A \subseteq X$.

Proof. ($\implies$) Note that $\overline{f(A)}$ is a closed set in Y and so its preimage under f , $f^{-1}(\overline{f(A)})$, is closed in X from the previous part. Since $f(A) \subseteq \overline{f(A)}$, we have that $A \subseteq f^{-1}(\overline{f(A)})$. Since $f^{-1}(\overline{f(A)})$ is a closed set in X containing A and $\overline{A}$ is the smallest closed set in X containing A , we have that $\overline{A} \subseteq f^{-1}(\overline{f(A)})$, meaning that $f(\overline{A}) \subseteq \overline{f(A)}$.

($\impliedby$) Let C be a closed set in Y . Our goal is to show that $f^{-1}(C)$ is closed in X ; that is, we want to show that $\overline{f^{-1}(C)} = f^{-1}(C)$. Notice that $f^{-1}(C) \subseteq \overline{f^{-1}(C)}$ is immediate. So, it suffices to show that $\overline{f^{-1}(C)} \subseteq f^{-1}(C)$. Indeed, by setting $A = f^{-1}(C)$ which is a subset of X and using our assumption, we get that

$$\begin{aligned} f(\overline{A}) \subseteq \overline{f(A)} &\implies \overline{A} \subseteq f^{-1}(\overline{f(A)}) \\ &\implies \overline{f^{-1}(C)} \subseteq f^{-1}[\overline{f(f^{-1}(C))}] \\ &\implies \overline{f^{-1}(C)} \subseteq f^{-1}(C). \end{aligned}$$

Thus, we conclude that $f^{-1}(C)$ is closed in X . ■

- (iii) Prove that if, f is continuous and A is dense in X , then $f(A)$ is dense in $f(X)$.

Proof. Our goal is to show that $\overline{f(A)} = f(X)$; that is, we want to show that $f(X) \subseteq \overline{f(A)}$ and $\overline{f(A)} \subseteq f(X)$. Starting with the first inclusion, we observe that since f is continuous and A is dense in X (i.e. $\overline{A} = X$), it follows from part (ii) that

$$f(X) = f(\overline{A}) \subseteq \overline{f(A)} \implies f(X) \subseteq \overline{f(A)}. \quad (1)$$

Now, we need to show that $\overline{f(A)} \subseteq f(X)$. Since $f^{-1}(\overline{f(A)}) \subseteq X$. Taking the image under f , it follows that $\overline{f(A)} \subseteq f(X)$ (2).

From (1) and (2), it follows that $\overline{f(A)} = f(X)$. ■

Problem 2 (Continuity and Lower Limit Topology). (i) Let $\mathcal{T}_L$ be the lower limit topology and $\mathcal{T}_E$ be the Euclidean topology on $\mathbb{R}$. Prove that $f : (\mathbb{R}, \mathcal{T}_L) \rightarrow (\mathbb{R}, \mathcal{T}_E)$ is continuous if and only if for every $x \in \mathbb{R}$ and $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all $y \in [x, x + \delta)$.

Proof. ($\implies$) Suppose f is continuous. Our goal is to show that for all $x \in \mathbb{R}$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all $y \in [x, x + \delta)$; that is, we need to show that

for all $x \in \mathbb{R}$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$[x, x + \delta] \subseteq f^{-1}(B_\varepsilon(f(x))).$$

Let $x \in \mathbb{R}$ and $\varepsilon > 0$ be given. Now, consider the open ball $B_\varepsilon(f(x))$ and notice that $x \in f^{-1}(B_\varepsilon(f(x)))$. Since $B_\varepsilon(f(x))$ is an open set in $(\mathbb{R}, \mathcal{T}_E)$ and f is continuous, we know that $f^{-1}(B_\varepsilon(f(x)))$ is open in $(\mathbb{R}, \mathcal{T}_L)$. Since $x \in f^{-1}(B_\varepsilon(f(x)))$, there exists an open set $U' \in \mathcal{T}_L$ containing x such that $x \in U' \subseteq f^{-1}(B_\varepsilon(f(x)))$. Since $U' \in \mathcal{T}_L$, we know that $U' = [a, b)$ for some $a, b \in \mathbb{R}$. Since $x \in U'$, define $\delta = \frac{1}{2}(b - x)$ (this is strictly positive because $x \in [a, b)$; that is, $a \leq x < b \implies b - x > 0$). Thus, we see that $[x, x + \delta] \subseteq U' \subseteq f^{-1}(B_\varepsilon(f(x)))$ and so we conclude that

$$[x, x + \delta] \subseteq f^{-1}(B_\varepsilon(f(x))).$$

($\Leftarrow$) Conversely, we will show that f is continuous. Suppose that for all $x \in \mathbb{R}$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that $[x, x + \delta] \subseteq f^{-1}(B_\varepsilon(f(x)))$. To show that f is continuous, we need to show that for all open sets U in $(\mathbb{R}, \mathcal{T}_E)$, $f^{-1}(U)$ is open in $(\mathbb{R}, \mathcal{T}_L)$.

To this end, let U be an open set with respect to $\mathcal{T}_E$. Let $x \in f^{-1}(U)$ be arbitrary. Then $f(x) \in U$, by definition. Since U is open in $\mathcal{T}_E$, there exists an $\varepsilon > 0$ such that $B_\varepsilon(f(x)) \subseteq U$ with respect to the standard metric on $\mathbb{R}$. By assumption, since $x \in \mathbb{R}$ and $\varepsilon > 0$,

$$\text{there exists a } \delta > 0 \text{ such that } [x, x + \delta] \subseteq f^{-1}(B_\varepsilon(f(x))).$$

Since $[x, x + \delta] \in \mathcal{T}_L$ ($[x, x + \delta]$ is open with respect to the Lower Limit Topology) containing x , it follows that

$$x \in [x, x + \delta] \subseteq f^{-1}(B_\varepsilon(f(x))) \subseteq f^{-1}(U) \implies x \in [x, x + \delta] \subseteq f^{-1}(U).$$

Thus, this implies that $f^{-1}(U)$ is open in $\mathcal{T}_L$ and so f is continuous. ■

- (ii) Is $f : (\mathbb{R}, \mathcal{T}_L) \rightarrow (\mathbb{R}, \mathcal{T}_E)$, $f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$ continuous?

Solution. We claim that f is continuous. Notice that f is continuous for all $x \in \mathbb{R}$ that is non-zero. It suffices to check that f is continuous at $0 \in \mathbb{R}$. We can see this because for all $\varepsilon > 0$, we can define $\delta = 1$ such that $[0, 1] \subseteq f^{-1}(B_\varepsilon(f(0))) = f^{-1}(B_\varepsilon(1))$. That is, f is continuous at 0. ■

- (iii) Is $f : (\mathbb{R}, \mathcal{T}_E) \rightarrow (\mathbb{R}, \mathcal{T}_L)$, $f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$ continuous?

Solution. No, because there does not exist an open set U in $\mathcal{T}_L$ such that under the preimage of f , $f^{-1}(U)$ is not open in $\mathcal{T}_E$. For example, take $U = [0, \varepsilon)$ for any $\varepsilon > 0$ in $\mathcal{T}_L$. Then $f^{-1}(U)$ is not open because $[0, \varepsilon)$ is not open with respect to the Euclidean topology. ■

Problem 3 (Pull-back topology). Let X be a set, $(Y, \mathcal{T})$ be a topological space, and $f : X \rightarrow Y$ be a function.

- (i) Prove that $f^*\mathcal{T} = \{f^{-1}(U) : U \in \mathcal{T}\}$ is a topology on X . (This is called the pull-back topology on X induced by f .)

Proof. Our goal is to show the following conditions:

- (i) $\emptyset, X \in f^*\mathcal{T}$;
 - (ii) For all $A_1, A_2 \in f^*\mathcal{T}$, $A_1 \cap A_2 \in f^*\mathcal{T}$;
 - (iii) If $\{A_\alpha\}_{\alpha \in \Omega}$ is an arbitrary collection in $f^*\mathcal{T}$, then $\bigcup_{\alpha \in \Omega} A_\alpha \in f^*\mathcal{T}$.
- (i) Notice that $f^{-1}(\emptyset) = \emptyset$ and $f^{-1}(Y) = X$ where $\emptyset, Y \in \mathcal{T}$. Thus, it follows that $\emptyset, X \in f^*\mathcal{T}$.

(ii) Let $A_1, A_2 \in f^*\mathcal{T}$. Then by definition, we have $A_1 = f^{-1}(U_1)$ and $A_2 = f^{-1}(U_2)$ for some U_1 and U_2 in $\mathcal{T}$. Then

$$A_1 \cap A_2 = f^{-1}(U_1) \cap f^{-1}(U_2) = f^{-1}(U_1 \cap U_2).$$

Notice that since U_1 and U_2 are in $\mathcal{T}$ and $\mathcal{T}$ is a topology on Y , it follows that $U_1 \cap U_2 \in \mathcal{T}$. Thus, it follows that $A_1 \cap A_2 \in f^*\mathcal{T}$.

(iii) Let $\{A_\alpha\}_{\alpha \in \Omega}$ be a collection of sets in $f^*\mathcal{T}$. Then for all $\alpha \in \Omega$, $A_\alpha = f^{-1}(U_\alpha)$ for some $U_\alpha \in \mathcal{T}$. Then

$$\bigcup_{\alpha \in \Omega} A_\alpha = \bigcup_{\alpha \in \Omega} f^{-1}(U_\alpha) = f^{-1}\left(\bigcup_{\alpha \in \Omega} U_\alpha\right).$$

Since $\{U_\alpha\}_{\alpha \in \Omega} \subseteq \mathcal{T}$ and $\mathcal{T}$ is a topology, it follows that $\bigcup_{\alpha \in \Omega} U_\alpha \in \mathcal{T}$. This tells us that

$$\bigcup_{\alpha \in \Omega} A_\alpha \in f^*\mathcal{T}.$$

Thus, from (i) through (iii), we conclude that $f^*\mathcal{T}$ is, indeed, a topology on X . ■

(ii) Prove that $f : (X, f^*\mathcal{T}) \rightarrow (Y, \mathcal{T})$ is continuous.

Proof. Let U be an open set in $(Y, \mathcal{T})$ and let $x \in f^{-1}(U)$ and so, $f(x) \in U$. Since U is open in $(Y, \mathcal{T})$, there exists an open set W in $(Y, \mathcal{T})$ containing $f(x)$ such that $f(x) \in W \subseteq U$. But this tells us that $x \in f^{-1}(W) \subseteq f^{-1}(U)$ where $f^{-1}(W)$ is open in $(X, f^*\mathcal{T})$. Hence, it follows that $f^{-1}(U)$ is an open set in $(X, f^*\mathcal{T})$. Thus, f is continuous. ■

(iii) Prove that, if S is a topology on X such that $f : (X, S) \rightarrow (Y, \mathcal{T})$ is continuous, then $f^*\mathcal{T} \subseteq S$.

Proof. Let $A \in f^*\mathcal{T}$ be arbitrary. Then it follows by definition that there exists an open set U in $(Y, \mathcal{T})$ such that $A = f^{-1}(U)$. Since $f : (X, S) \rightarrow (Y, \mathcal{T})$ is continuous and U is an open set in $(Y, \mathcal{T})$, $f^{-1}(U)$ is open in (X, S) , but this tells us that A is open with respect to S in X ; that is, $A \in S$. Hence, $f^*\mathcal{T} \subseteq S$. ■

Problem 4 (Open and Closed Sets Via Continuous Functions). (i) Prove that $\{x \in \mathbb{R} : \cos x > \sin x\}$ is open in $\mathbb{R}$.

Proof. Define $f(x) = \cos x - \sin x$ where $f : \mathbb{R} \rightarrow \mathbb{R}$. Since $\sin x$ and $\cos x$ are continuous on $\mathbb{R}$ and that the difference of continuous functions is continuous, it follows that $f(x)$ is continuous. Consider the open set $(0, +\infty)$ in $\mathbb{R}$. We see that

$$\begin{aligned} f^{-1}(0, +\infty) &= \{x \in \mathbb{R} : f(x) \in (0, +\infty)\} \\ &= \{x \in \mathbb{R} : f(x) > 0\} \\ &= \{x \in \mathbb{R} : \cos x - \sin x > 0\} \\ &= \{x \in \mathbb{R} : \cos x > \sin x\}. \end{aligned}$$

Since f is continuous and $(0, +\infty)$ is open in $\mathbb{R}$, we know that $f^{-1}((0, +\infty))$ is open in $\mathbb{R}$. That is, $x \in \mathbb{R} : \cos x > \sin x$ is open in $\mathbb{R}$. ■

(ii) Show that $\{(x, y) \in \mathbb{R}^2 : x > y, x^2 + y^2 < 1\}$ is open in $\mathbb{R}^2$.

Proof. Define $f(x, y) = x^2 + y^2$ where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Notice that f is continuous because x^2 and y^2 are polynomials in $\mathbb{R}$ which are continuous and that the sum of these two continuous functions is continuous. Thus, f is continuous on $\mathbb{R}^2$. Now, consider the open set $(-\infty, 1)$ in

$\mathbb{R}$. Since f is continuous, we know that its preimage $f^{-1}(-\infty, 1)$ is open in $\mathbb{R}^2$. Then

$$\begin{aligned} f^{-1}(-\infty, 1) &= \{(x, y) \in \mathbb{R}^2 : f(x, y) \in (-\infty, 1)\} \\ &= \{(x, y) \in \mathbb{R}^2 : f(x, y) < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}. \end{aligned}$$

Hence, $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$ is open in $\mathbb{R}^2$. ■

(iii) Prove that $\{(x, y, z) \in \mathbb{R}^3 : 4 \leq x^2 + y^2 + z^2 \leq 6\}$ is closed in $\mathbb{R}^3$.

Proof. Define $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ by $f(x, y, z) = x^2 + y^2 + z^2$. Since x^2, y^2 , and z^2 are polynomials in $\mathbb{R}$ which are continuous, it follows from the fact that a sum of continuous functions that f is continuous. Consider the closed set $[4, 6]$ in $\mathbb{R}$. Since f is continuous, its preimage $f^{-1}([4, 6])$ is closed in $\mathbb{R}^3$. That is,

$$\begin{aligned} f^{-1}([4, 6]) &= \{(x, y, z) \in \mathbb{R}^3 : f(x, y, z) \in [4, 6]\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : 4 \leq f(x, y, z) \leq 6\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : 4 \leq x^2 + y^2 + z^2 \leq 6\}. \end{aligned}$$

Hence, it follows that $\{(x, y, z) \in \mathbb{R}^3 : 4 \leq x^2 + y^2 + z^2 \leq 6\}$ is closed in $\mathbb{R}^3$. ■

Problem 5 (Pasting Lemma For A Finite Cover). Assume the hypothesis of the pasting lemma. Show that if A_i is closed for all $i \in I$ and I is finite, then f is continuous.

Proof. Let U be closed in Y and let $\{A_i\}_{i \in I}$ be a finite cover of X . Our goal is to show that $f^{-1}(U)$ is open in X . Notice that

$$\begin{aligned} f^{-1}(U) &= f^{-1}(U) \cap X = f^{-1}(U) \cap \left(\bigcup_{i=1}^n A_i \right) \\ &= \bigcup_{i=1}^n [f^{-1}(U) \cap A_i]. \end{aligned}$$

Note that since $f(x) = f_i(x)$ for all $x \in A_i$, we get that for all $1 \leq i \leq n$

$$\begin{aligned} f_i^{-1}(U) &= \{x \in A_i : f_i(x) \in U\} \\ &= \{x \in A_i : f(x) \in U\} \\ &= \{x \in A_i : x \in f^{-1}(U)\} \\ &= A_i \cap f^{-1}(U). \end{aligned}$$

Then we get that

$$f^{-1}(U) = \bigcup_{i=1}^n f_i^{-1}(U).$$

Since each f_i is continuous on A_i for all $1 \leq i \leq n$, and U is a closed set in Y , it follows that $f_i^{-1}(U)$ is closed for all $1 \leq i \leq n$. Since a finite union of closed sets is closed, it follows that

$$f^{-1}(U) = \bigcup_{i=1}^n f_i^{-1}(U)$$

is a closed set and so we conclude that f is continuous. ■

Problem 6. (i) Let X be a topological space and A is a subspace of X . Consider the inclusion $i : A \rightarrow X$, $i(x) = x$. Then

(a) A is open in X if and only if the inclusion i is open.

Proof. Let $\mathcal{T}$ be a topological space on X and let $\mathcal{T}|_A$ be the subspace topology endowed from X . ($\implies$) Suppose A is open. Our goal is to show that the inclusion map i is open; that is, $i(U) \in \mathcal{T}$ for all $U \in \mathcal{T}|_A$. Let $U \in \mathcal{T}|_A$ be arbitrary (that is, U is open in A). We want to show that $i(U)$ is open in X . To this end, let $y \in i(U)$ be arbitrary. Then there exists an $x \in U$ such that $i(x) = y$. But U is open in A , so there exists an open set V in X such that $U = V \cap A$. But A is open in X , so we have that U is open in X . Since $x \in U$ and U is open in X , there exists an open set W in X containing x such that $x \in W \subseteq U$. Since $i(x) = y$, by definition of the inclusion map, $i(x) = x$ and so $x = y$. Now, note that $W \subseteq U$ implies that $i(W) \subseteq i(U)$ since $x \in W$. However, $i(W) = W$ since W is open in X , and so we get that

$$i(x) \in i(W) \subseteq i(U) \implies y \in W \subseteq i(U).$$

This tells us that $i(U)$ is open in X and so i is open.

($\impliedby$) Suppose that i is open. Then i is open if and only if for all $U \in \mathcal{T}|_A$, $i(U) \in \mathcal{T}$. In particular, $A \in \mathcal{T}|_A$, and so $i(A) \in \mathcal{T}$. But note that $i(A) = A$ and so $A \in \mathcal{T}$. Thus, it follows that A is open in X . ■

(b) A is closed in X if and only if the inclusion i is closed.

Proof. ($\implies$) Suppose A is closed in X . Our goal is to show that i is closed; that is, we need to show that for all closed sets V in A , $i(V)$ is closed in X . To this end, let V be a closed set in A . Then there exists a closed set U in X such that $V = U \cap A$ is closed in X (since the finite intersection of two closed sets is closed). This tells us that V^c is open in X . Set $W = V^c$. Then i is open from part (a); that is, for all $B \in \mathcal{T}|_W$, $i(B) \in \mathcal{T}$. Since $W \in \mathcal{T}|_W$, it follows that $i(W) \in \mathcal{T}$. Then $i(V^c) \in \mathcal{T}$ and so $[i(V)]^c \in \mathcal{T}$. That is, $i(V)$ is closed in X .

($\impliedby$) Suppose that i is closed. That is, for all closed sets V in A , $i(V)$ is closed in X . Since A itself is closed in its subspace topology, it follows that $i(A)$ is closed in X . But by definition $i(A) = A$ and so A is closed in X . ■

(ii) Let $n, k \in \mathbb{N}$. Prove that the function

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^{n+k} \quad f(x_1, x_2, \dots, x_n) = (x_1, x_2, \dots, x_n, 0, 0, \dots, 0)$$

is not open.

Proof. Construct the open ball $B_\varepsilon(0)$ centered at 0 with an arbitrary radius ε in $\mathbb{R}^n$. The image of this ball under f is

$$f(B_\varepsilon(0)) = \{(x_1, x_2, \dots, x_n, 0, 0, \dots, 0) : x \in B_\varepsilon(0)\}.$$

Note that there does not exist an open ball in $\mathbb{R}^{n+k}$ containing 0 that is completely contained in the above image. We can see this by taking any point with k -th component that is non-zero. It will be contained in the open ball, but not in the image because one of the k th components past n is non-zero. Hence, f is not an open map. ■